Auditing Contextual Inverse Optimization across Problem Families

Identifiability, optimal-face sensitivity, and a frozen local evaluation

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Abstract

Contextual inverse optimization seeks to infer missing optimization costs from descriptions and observed decisions. A small decision regret can nevertheless coexist with inaccurate parameters and unstable choices among estimated optima. We investigate these issues using a preserved historical diet audit and a prospectively frozen extension to inventory relocation and symmetric traveling-salesperson problems from MAMO. Of 59 candidate source problems, 19 satisfy the implemented reconstruction and answer-consistency checks; 16 are evaluation problems and 3 are development problems. Five context seeds, two history counts, two noise caps, and eight local methods produce 3,040 recorded runs without solver failures. At the primary setting of 20 histories and a 10% suboptimality cap, text-selected sparse inverse optimization has mean regret 5.540% on 11 relocation problems and 0.764% on 5 tour problems. Fixed midpoint costs yield 8.118% and 4.271%, respectively. Source-paired exploratory intervals cross zero for relocation but not for tours. Crucially, the relocation method's worst estimated-optimal-face mean regret is 318.944%, despite its favorable default-solver mean. A positive-price prior removes this large face ambiguity in the measured cohort. These findings support further study of identification and solution selection, not general claims of language-model understanding, universal superiority, or reliable cost recovery.

1. Motivation and scope

Operational models often omit costs that practitioners nevertheless incorporate into their decisions. Text can suggest the missing terms, while historical decisions may restrict their weights. This motivates a three-stage system: generate a parameterized structural hypothesis, solve an inverse optimization problem, and refine the structure using residuals and regularization. The present study isolates the mathematical calibration and evaluation stages. It uses controlled textual templates to select candidate costs, rather than asking a paid language model to generate hypotheses. This makes computation inexpensive and reproducible while deliberately limiting what can be concluded about the full proposed system.

The central question is not simply whether an inverse solver improves an average score. We ask whether improvement survives problem-level repetition, whether inferred costs are identified by the histories, and whether arbitrary choices among estimated-optimal solutions affect the conclusion. Our contribution is an evidence package combining a frozen cross-family protocol, complete run retention, two non-learning controls, six local learning or diagnostic methods, and explicit optimal-face witnesses. This is an exploratory software research draft, not a claim of a new inverse optimization theorem or a conference-ready submission.

The earlier diet experiment is retained unchanged as historical evidence. Its audit found that two sources dominated mean regret, some parameters were never activated, and changing solver variable order changed default decisions without changing the mathematical model. The extension was designed in response to those findings, but new results are not substituted into old tables. Experimental phases, test contexts, and metrics remain identifiable.

2. Related work and attribution

Chan, Mahmood, and Zhu [1] survey inverse optimization theory and applications, providing the conceptual distinction between inferring an objective and reconstructing constraints. Mohajerin Esfahani et al. [2] study

imperfect information and data-driven inverse optimization, including tractable formulations based on suboptimality. Our objective uses a suboptimality residual, but we do not implement their distributionally robust formulation or inherit its statistical guarantees merely by using a related loss.

Kitaoka [3] develops a gradient-based approach with finite exact-solution properties for a specified data-driven inverse MILP problem. The projected subgradient control in this study is a fixed-budget, elementary implementation. It is not a reproduction of that paper and carries no finite exactness or parameter recovery guarantee. Algorithmic finite attainment must not be interpreted as a theorem that a few noisy histories identify every unknown price.

MAMO [4] supplies the public natural-language optimization problems used here. We use a pinned repository revision and inspect the explicit numerical models. Generated context variants and near-optimal decisions are our controlled additions, not human decisions released by MAMO. The documented disagreement between some parsed tour problems and public answers motivates transparent admission records and independent optimization checks. It does not establish that all MAMO problems are wrong.

3. Parameterized models and the inverse objective

For context k, let X_k be the feasible set, b_k the known objective coefficients, F the candidate feature matrix, and theta a nonnegative vector of unknown costs. The forward model is P_k(theta): minimize (b_k + F theta)^T x over x in X_k. Two entries of theta represent masked true costs. Three further entries represent candidate additional charges on other arcs or edges; their oracle weights are zero. A controlled template selects the two true entries and one distractor. The learner receives the feasible model, known coefficients, selected features, and observed decisions. Oracle coefficients and oracle test solutions are excluded from learner inputs.

For a feasible observation xhat_k, define loss_k(theta) = (b_k + F theta)^T xhat_k - min_{x in X_k}(b_k + F theta)^T x. It is nonnegative and convex in theta for a fixed feasible set, including a finite mixed-integer feasible set, because it is a pointwise maximum of affine functions. The main estimator minimizes mean(loss_k)/s + lambda * sum_j(theta_j)/U subject to 0 <= theta_j <= U for selected terms and theta_j = 0 otherwise. Here s = max(1, mean_k b_k^T xhat_k), lambda = 0.01, and U is a declared family-specific bound: 60 for relocation, 100 for tours.

An epigraph variable e_k is constrained by e_k >= b_k^T(xhat_k-x) + theta^T F^T(xhat_k-x) for each feasible competitor x. The implementation alternates a linear master solve with forward optimization to find violated inequalities. This avoids treating integer forward models as though they possessed an ordinary continuous LP dual. Relocation forward problems are continuous linear programs; tour competitors are found by complete cycle enumeration. The master stops when no residual exceeds its epigraph value by more than 1e-6, with a maximum of 100 separation rounds. HiGHS is single-threaded with a 10-second per-solve time limit and zero requested MIP gap.

The implementation does not infer arbitrary hidden constraints or a full coefficient matrix. Candidate structure is restricted to five unit-cost terms. Claims about implicit-constraint F1, general MILP reconstruction, or open-ended textual hypothesis generation therefore require separate experiments. We do not report a structure-recovery score for a task that did not actually discover new feasible-set constraints.

4. Identification and solution-selection diagnostics

Observation 1 (inactive nonnegative feature). Suppose an observed decision is optimal for some cost vector, a candidate feature is nonnegative on all feasible decisions, and that feature equals zero at every observation. Increasing its cost preserves optimality of every observation: the observed objective is unchanged while every competitor becomes no cheaper. Thus the data do not imply a finite upper bound without additional restrictions. In noisy suboptimality fitting, increasing this cost can likewise leave or reduce the residual. A box bound or regularizer chooses a value but is not itself evidence of data identification. This elementary argument explains why an unused arc price can have large error without harming observed fit.

Observation 2 (zero-cost ambiguity). If known cost is inactive on observations and all selected unknown costs can be zero, the inferred objective can make many feasible decisions equally optimal. Nonnegative L1 regularization favors zero where it decreases the penalized objective. A known objective component elsewhere in the model does not automatically anchor scale on the observed choices. This is a structural failure mode, not necessarily solver randomness.

For an estimated optimum vhat_k, we define an optimal-face diagnostic set by x in X_k and chat_k^T x <= vhat_k + 1e-7. We separately minimize and maximize oracle cost on this set. After subtracting the oracle optimum and normalizing, these give best and worst regret among estimated optima. An unbounded maximum is retained as unbounded, rather than converted to a finite favorable score. The diagnostic uses oracle information and cannot be used to pick a deployment decision. Default-solver regret remains the primary metric, preserving comparability with historical reports.

Across-seed variation and within-face variation answer different questions. Five context seeds assess sensitivity to generated training and test conditions. The optimal-face range assesses ambiguity even with a fixed estimate and context. Neither range is a confidence interval. A post-outcome witness was also solved with an exact estimated-cost cap and a tighter feasibility tolerance to test whether a large face range was only numerical slack.

5. Data provenance and admission

The source is the MAMO repository at commit 2c91069dee5c1e4db4b899eb176ec987d0251bb3, file mamo_complex_lp.jsonl. We audited source IDs 41-59 as relocation candidates and 60-99 as tour candidates. Before inverse outcomes, admission required a complete explicit numerical model supported by the adapter, compatible semantics, and a reconstructed optimum matching the public answer within 1e-6. Exact numerical duplicates would be rejected. Sources with IDs divisible by five were assigned to development, and all other admitted sources to evaluation.

Fourteen of nineteen relocation candidates were admitted. One excluded source used exact-final-inventory language inconsistent with total excess inventory without specified disposal; four could not be completely parsed by the current stock/need or cost adapter. Such parser exclusions are adapter limitations, not automatic evidence of invalid source data. The admitted source IDs are 41, 42, 43, 44, 45, 46, 47, 50, 51, 52, 53, 54, 55, and 59. Development IDs are 45, 50, and 55, leaving 11 evaluation sources.

Thirty-four of forty tour candidates were fully parsed. Twenty-nine reconstructed closed-tour optima disagreed with the public answers; six other candidates could not be fully parsed. The five admitted evaluation sources are 64, 68, 74, 84, and 87. Every one of the 34 parsed models was also solved by an independent directed MTZ integer formulation, agreeing with enumeration. These checks share the text parser and HiGHS solver, so they do not constitute independent human adjudication of every statement. For example, source 61 has reconstructed closed-tour optimum 121, while its public answer is 60.

Admission was frozen before inverse evaluation, rather than changed after observing scores. Nonetheless, retaining only answer-consistent, supported problems limits coverage and can induce selection effects. Nineteen admitted sources are not a representative random sample of the full benchmark. Family-specific conclusions must be restricted to this cohort, especially the five tour problems. Complete rejected-source records accompany the release.

6. Contexts, histories, and frozen protocol

Each admitted source masks two fixed true costs and samples three distractor positions using a source-specific deterministic seed. For each of seeds 11, 23, 37, 53, and 71, twenty training and twenty test contexts are generated. Known positive costs are independently multiplied by Uniform(0.5,1.5), while the hidden true costs remain fixed. Relocation net demand balances are multiplied by a common Uniform(0.8,1.2) factor and rounded, with feasibility checked. Tour feasible cycles remain fixed. These context distributions are controlled synthetic perturbations of real published problem statements.

Relocation decisions are nonnegative continuous flows. Initial inventory plus inflow minus outflow must meet each location's need, excess stock may remain, and transshipment is permitted. Artificial flow upper bounds are not added to suppress ambiguity. Tours are undirected Hamiltonian cycles on four to six cities, visiting every city and returning to the start. Enumeration removes reversal duplicates but retains every feasible cycle.

Training observations are either certified optimal decisions or feasible decisions within a 10% oracle-cost gap cap. For noisy histories, a fixed positive auxiliary objective is optimized subject to that cap. The auxiliary seed is coupled across conditions; zero noise uses an oracle optimum. This is a bounded-suboptimality generator, not an empirical model of human mistakes. Ten percent is a cost-gap bound, not a probability of corruption. Both K=5 and K=20 are evaluated, with K=5 taking the first five of the same twenty histories.

The primary setting is K=20 and noise cap 0.10. Every method shares the same contexts, features, masks, and histories. Inputs, protocol, and experiment implementation were hashed before the inverse runs. No hyperparameter was selected using test scores. Three development sources are retained in raw results but excluded from every comparative summary. The design contains 19 sources x 5 seeds x 2 history counts x 2 noise caps x 8 methods = 3,040 scheduled rows. Old diet contexts and scores remain separate.

7. Controls, comparisons, and their limits

Fixed midpoint uses [U/2,U/2,0,0,0] without history learning. Known-cost median fills the two missing costs with the median of positive known training-context costs and sets distractors to zero. These are essential non-learning controls. Correct-support IO estimates only the two genuine missing terms, without L1 regularization; it is an oracle-support diagnostic and has information unavailable in an open structural discovery task.

Text + sparse IO estimates the controlled template's three terms using lambda=0.01. Full-bank IO uses all five terms with the same penalty, providing a restricted candidate-filter comparison rather than full-matrix inverse optimization. Positive-floor IO adds theta_1,theta_2 >= 1 to text + sparse IO, leaving nuisance lower bounds at zero. The floor is an explicit positive-integer-price domain prior supported by these original costs. It is not inferred from observations and cannot silently be carried to domains allowing zero or fractional true prices.

Prior-center IO first minimizes unregularized training loss on the selected three terms, then minimizes L1 distance to [U/2,U/2,0] among fits within 1e-6 normalized loss of that minimum. Projected subgradient starts at the same center, runs 100 cyclic single-history updates with step U/sqrt(t)/max(1,gradient norm), projects to the nonnegative box and selected features, and returns the best full-training-loss checkpoint among initialization and every tenth step. It has no claimed exactness guarantee and is not tuned on test outcomes.

Only specific contrasts isolate one factor. Text-selected versus full-bank sparse IO changes the candidate set at fixed penalty. Positive-floor versus text sparse IO changes the lower-bound prior. Correct-support versus text sparse IO changes both support and regularization, so it is not a clean sparsity ablation. This phase contains no same-support, no-L1 paired control; it cannot separately establish a benefit of sparsity. We publish all eight methods and all conditions rather than elevating the observed best method into a prospectively designated winner.

8. Metrics and uncertainty

For a test context, regret is max(0, ctrue^T xpred - voracle)/max(1, abs(voracle)). A run averages this quantity over twenty test contexts. A source-level score averages its five runs; family means weight sources equally. Hidden-cost MAE averages absolute error on the two true hidden entries, in that family's original cost units. We do not pool heterogeneous cost units across families. Decision quality and cost recovery are reported separately because equivalent decisions can arise from different prices.

For each source and condition, the release includes the minimum, median, maximum, and mean across five seeds. Family tables also include the median across source means and the range across all runs, with explicit labels. Pairwise differences are computed within source after averaging seeds, followed by 5,000 source-level bootstrap resamples with seed 20261021. These are exploratory percentile intervals without multiplicity correction. They are not confirmatory p-values, and five tour sources cannot support broad population inference.

Repeated contexts do not turn a source into many independent benchmark problems.

No failed row is dropped or favorably imputed. No failures occurred in this run. If failures occur on reproduction, a complete source mean is withheld rather than computed only from its successful seeds. All test decisions, solver status labels, predicted optima, optimal-face costs, parameter estimates, and training losses where available are retained in the JSON release. The reported verification count is a count of program assertions, not an independent experimental sample size.

9. Results and counterexamples

At the primary condition, tour text + sparse IO obtains 0.764% mean regret versus 4.271% for fixed midpoint, a paired difference of -3.507 percentage points with exploratory interval [-5.139,-1.822]. All five source-averaged differences favor the text-selected estimator. Prior-center IO and projected subgradient have observed means 0.423% and 0.431%. These observations motivate further tests on genuinely independent sources; they do not rank algorithms conclusively across the broader class of routing problems.

Relocation text + sparse IO obtains 5.540% versus 8.118% for midpoint, with difference -2.578 percentage points and interval [-13.370,+7.093]. It wins on four sources, ties on two, and loses on five. Projected subgradient has the lowest observed mean, 1.311%, but its difference interval against midpoint is [-15.838,+0.056] percentage points. Thus a lower mean does not establish stable superiority. The mean and median differ substantially because a few conditions carry large losses.

Cost recovery exposes another limitation. Relocation text + sparse IO has hidden-cost MAE 20.106, compared with 11.545 for midpoint. Its average decisions improve while its absolute cost estimates worsen. In the primary relocation histories, 58 of 110 parameter-by-seed positions are never used, affecting nine of eleven sources; for tours, 13 of 50 positions are unused, affecting three of five sources. These activation counts do not replace a formal identified-set calculation, but they directly warn against interpreting low regret as accurate price recovery.

The strongest sensitivity witness is relocation source 44, seed 71, test context 24. The text-selected estimate assigns one hidden cost zero. A default plan has true cost 342.86, equal to the oracle optimum. Another feasible plan has the same estimated cost 342.86 but true cost 19,686.86, yielding 5,641.953% regret. The exact-cap tighter-tolerance witness confirms that this is not created by the 1e-7 diagnostic slack. Free inferred relocation can move surplus stock without changing estimated cost. The full default decision and the alternative witness are released for recalculation.

Across the relocation primary cohort, text + sparse IO's best-face and default means are both approximately 5.540%, but its worst-face mean is 318.944%. Positive-floor IO has default and worst-face means approximately 5.145%, removing the large observed ambiguity. This is a conditional success of a declared prior, not evidence that the original estimator learned positivity. Finite worst-face values in this cohort do not establish boundedness for every possible model; a unit test separately covers an unbounded zero-cost flow cycle.

The restricted candidate-filter comparison has modest, family-dependent effects. Full-bank minus text-selected mean regret is +0.591 percentage points for relocation, with exploratory interval [+0.003,+1.557], and +0.194 for tours, with interval [-0.007,+0.509]. These small-cohort comparisons are unadjusted and use a controlled template with true terms deliberately present. They cannot demonstrate free-form language understanding or justify claiming that textual priors always help.

Method	Relocation regret %	Tour regret %	Relocation MAE	Tour MAE
Fixed midpoint	8.118	4.271	11.545	27.400
Known-cost median	7.339	4.091	13.718	26.600
Correct-support IO*	4.655	0.807	19.992	18.463
Text + sparse IO	5.540	0.764	20.106	18.820
Full-bank IO	6.131	0.958	19.918	19.447
Positive-floor IO	5.145	0.764	19.696	18.838
Prior-center IO	2.821	0.423	9.815	16.770
Projected subgradient	1.311	0.431	9.536	15.747

Table 1. Primary K=20, gap cap=10%. Source-equal means after five seeds. MAE in original cost units. *Oracle-support diagnostic.

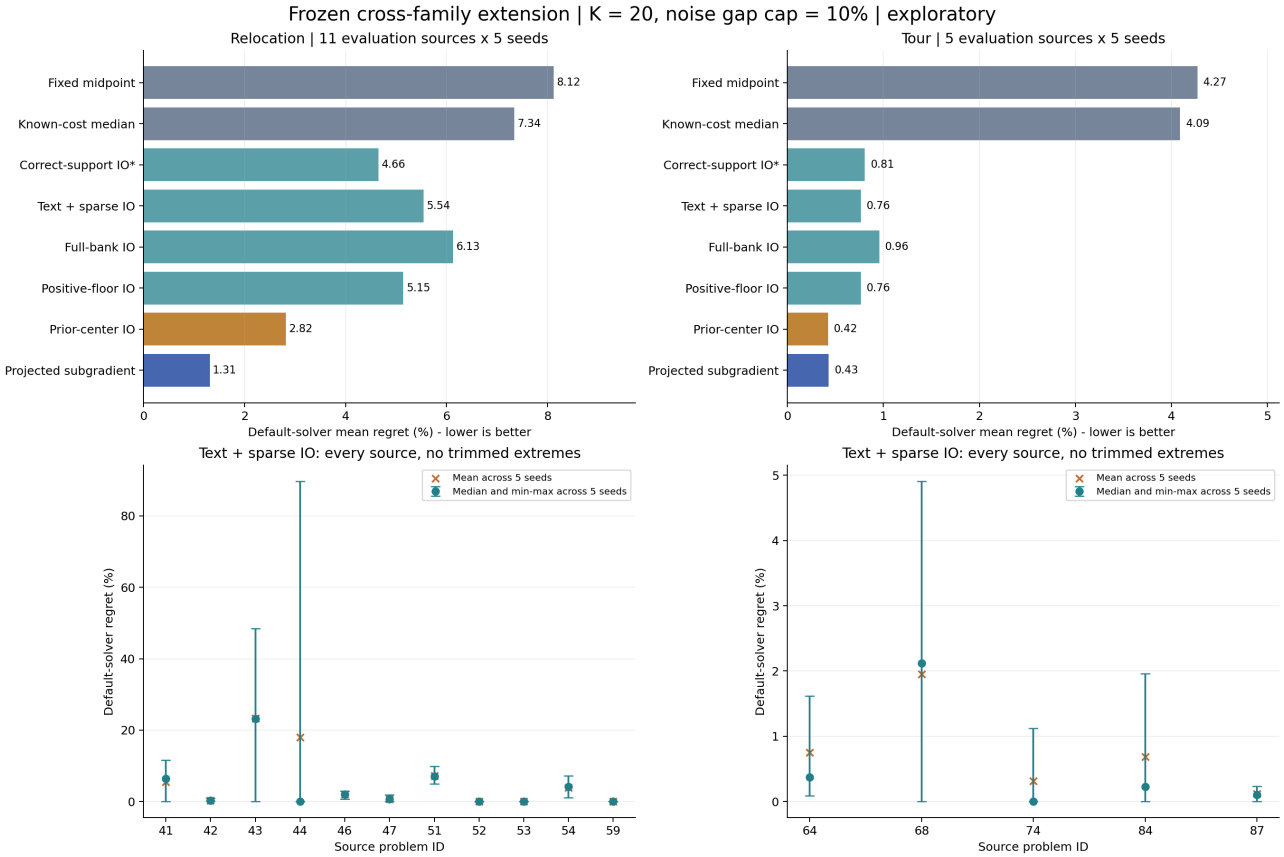


Figure 1. All methods and every evaluation source. Ranges show min/max over five seeds, not confidence intervals.

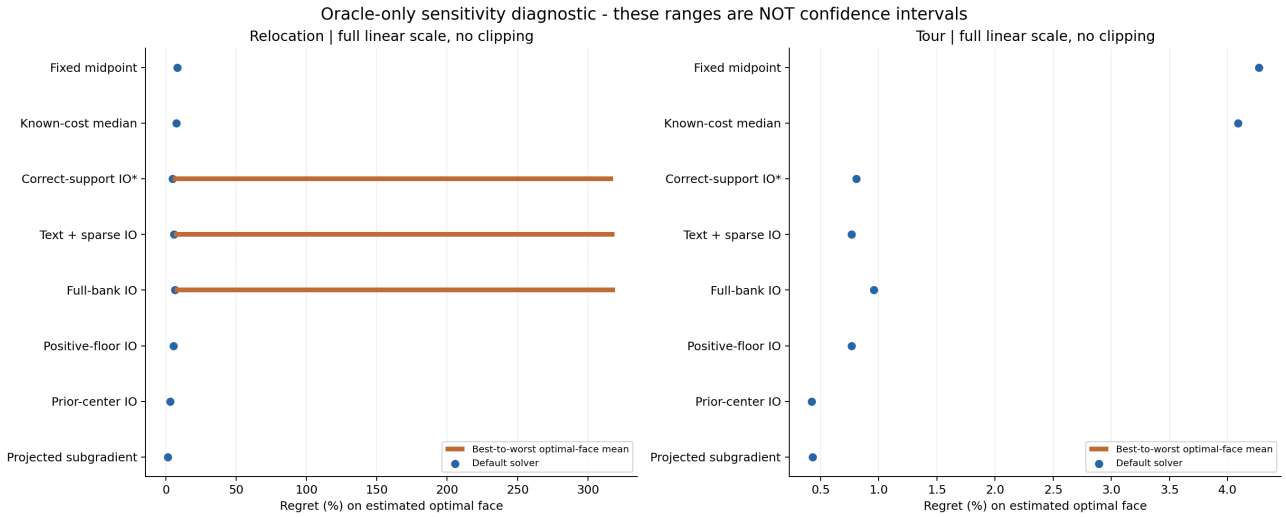


Figure 2. Oracle-only estimated-optimal-face sensitivity. Full linear axes retain large losses; panels have different scales.

10. Reproducibility and preserved evidence

The release includes the protocol, source admission ledger, frozen context files and hashes, all 3,040 result rows, all per-test predictions, analysis code, CSV summaries, figures, and this draft. A verifier checks 200,076 conditions including frozen-file hashes, complete unique schedule coverage, train/test separation, history feasibility and cost caps, predicted-decision feasibility, regret arithmetic, and inclusion of default regret in recorded face bounds. These checks validate records and arithmetic; they do not by themselves independently re-solve every optimization problem.

Independent formulation checks cover all 34 parsed tour sources. In addition, the complete source-44 seed-71 job was freshly re-solved for all eight methods, both K values, and both noise caps: all 32 rows were exactly equal as JSON to the stored run. The solver and library versions are recorded in freeze.json. All experiments in this extension use local computation and zero paid model calls. No human conversation logs, API credentials, or private operational records are part of the research artifact.

The previous diet report and its audit archive remain immutable historical evidence. The software version stays at 0.1.0, while experimental output directories carry phase identifiers. Reproduction instructions specify pinned-source acquisition, the local runner, the verifier, and the independent crosscheck. Scientific data derived from MAMO retain its CC BY-SA 4.0 attribution; this does not assert a new license over unrelated application code.

11. Limitations and next falsifiable tests

First, this phase adds two mathematical families but only one source dataset. Severe tour admission attrition and adapter exclusions limit external validity. Second, costs and histories are partly synthetic: the template is not a human log and the auxiliary near-optimal generator is not a calibrated human behavior model. Third, only two cost weights per source are hidden, and all constraints are given. Fourth, tours have at most six cities; exhaustive forward enumeration is practical here but does not establish scalability to industrial MILPs.

Fifth, five seeds measure sensitivity to the chosen generator and five tour sources provide weak evidence about new populations. Bootstrap intervals are exploratory and many comparisons are shown. Sixth, exact optimizer choices and numeric tolerances can influence predictions, which is why full face diagnostics are retained. Seventh, the price floor and center use prior knowledge that must be justified separately for each application. Eighth, no runtime benchmark or competitive language-model experiment is reported in this extension.

A subsequent protocol should validate adapters on an independent benchmark before inverse outcomes, use natural logs with blinded structural annotations, explicitly estimate parameter identified sets, and compare same-support regularized and unregularized methods. It should also include a faithful implementation of selected published baselines and a predeclared runtime budget. These are future falsifiable tests, not completed work.

The current results are sufficient to identify concrete mathematical risks and to motivate that next stage, not to claim a CCF A standard has already been met.

12. Conclusion

Cross-family evaluation reveals both encouraging decisions and unresolved identification. In a small admitted tour cohort, several local inverse methods improve on non-learning controls. Relocation gains are less stable, and default-optimum scores can hide extremely costly alternatives that the learned model also calls optimal. Explicit priors can help under their assumptions, while low regret alone cannot certify recovered prices. Preserving failed admissions, complete repeated measurements, and optimal-face witnesses gives a more defensible foundation for further contextual inverse optimization research.

References

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